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The University at Buffalo's response to Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

Introduction

The University at Buffalo (UB), a flagship campus of the State University of New York (SUNY) system, has long been a leader in artificial intelligence (AI) research. With a rich history of pioneering contributions from the early development of AI-powered handwriting recognition systems for postal automation[1] to leading the National AI Institute for Exceptional Education (AI4ExceptionalEd)[2] and the National Center for Early Literacy and Responsible AI (CELaRAI)[3], UB has consistently advanced AI technologies, specializing in cybersecurity, information security and biometrics, with a focus on addressing societally significant and challenging issues.

As the U.S. seeks to define and solidify its leadership in AI, it is essential to prioritize research and development that both drive economic growth and harness AI's transformative potential to address critical challenges in areas such as education, public safety and healthcare. UB's longstanding commitment to these areas of AI research offers important insights and recommendations for the development of an AI Action Plan. We propose policy priorities that will not only enhance U.S. leadership in AI but

[1] https://www.buffalo.edu/research/research-expertise/case-study-ai-at-usps.html
[2] https://www.nsf.gov/awardsearch/showAward?AWD_ID=2229873.
[3] https://ies.ed.gov/use-work/awards/center-early-literacy-and-responsible-ai-celarai-innovating-beginning-reading-instruction-culturally

also ensure that AI serves as a tool for public benefit, economic competitiveness, and global influence.

UB's Role in Pioneering AI Technologies

The University at Buffalo has been a leader in AI research for decades, contributing to advancements that have had both academic and societal significance. One of the earliest examples of UB's success in AI was the development of handwriting recognition technology in the 1990s, which revolutionized postal automation systems and became one of the first major practical success stories of AI by improving mail delivery while simultaneously generating significant savings of millions of dollars for the US Postal Service[4]. This early success demonstrated the immense potential of AI technologies to solve complex, real-world problems.

As AI technology has advanced, UB has continued to push the boundaries of innovation. More recently, UB received funding from the NSF and IES NCSER to establish the National AI Institute for Exceptional Education, spearheading the development of AI tools to screen, diagnose, and intervene for children with speech and language disorders. This multidisciplinary center represents a critical intersection of domains such as computer science, speech, language pathology, communication and education, with the potential to provide personalized, scalable interventions for children with developmental challenges. UB's research in this area underscores the potential for AI to dramatically improve educational outcomes and help children overcome obstacles to learning and communication across the country.

In addition to these groundbreaking projects, UB has continued to expand its AI research into other critical areas. These efforts contribute to the national and global dialogue on how AI can be used to address complex societal issues. Recently, UB has announced the creation of a new department, Department of AI and Society, with close collaboration between UB's School of Engineering and Applied Sciences (SEAS) and College of Arts and Sciences (CAS). UB's collaborations with industry, government, and other academic institutions will continue to ensure that its AI research leads to both academic advancements and real-world applications. These partnerships are essential in ensuring that AI innovation aligns with national priorities in healthcare, transportation, education, and sustainability, ensuring U.S. leadership in these areas of significant national interest.

Recommendations for the AI Action Plan: Policy Priorities for Conducting Use-inspired AI Research to Solve Societally Challenging Issues, Establishing U.S.

[4] https://www.govexec.com/federal-news/1999/02/postal-service-tests-handwriting-recognition-system/1746/

Leadership in Transforming Society with Positive Impact, and Boosting U.S. Global Competitiveness.

To maximize the societal impact of AI and maintain U.S. leadership in this critical area, we strongly endorse the following policy actions:

Priority #1: Accelerate Funding for AI Research with Societal Impact

The U.S. government should significantly increase funding for AI research focused on solving societal challenges, many of which are not a primary focus for commercial investment. Universities are well-positioned to lead research that addresses those critical issues. One such area is AI-augmented special education, which can transform the lives of children and greatly enhance their ability to contribute to the workforce and the economic prosperity of the nation.

Why it Enhances U.S. Competitiveness: Incentivizing the development of AI-driven solutions focused on societal challenges will improve the quality of life for the nation's citizens; and emphasizing AI for socially impactful initiatives is an investment for the long-term prosperity, security, and social well-being of the United States.

Priority #2: Create AI Innovation Hubs to address Societal Challenges

The U.S. should prioritize fostering partnerships between academic institutions, government agencies, and private industry to accelerate AI innovations. Universities are uniquely positioned to serve as hubs for such collaborations with industry, government and non-profit organizations. U.S. universities have traditionally been at the forefront of advancing the state of the art and partnerships with industry can help to quickly translate the research into practical, market-ready applications. To this end, the government should incentivize the creation of AI innovation hubs, where stakeholders from various sectors can collaborate to create AI solutions that have both commercial and societal value.

Why it Enhances U.S. Competitiveness: Public-private partnerships drive the rapid commercialization of AI technologies, positioning U.S. companies and institutions as global leaders in an AI-driven economy.

Priority #3: Promote AI Literacy and Workforce Development

To remain competitive in the AI space, the U.S. must invest heavily in AI education and workforce development. This includes integrating AI literacy into K-12 curricula, creating robust undergraduate and graduate programs focused on

AI, and offering reskilling programs for the existing workforce. Universities play a vital role in this critical endeavor. The government should also incentivize companies to invest in AI-related training programs that prepare workers for the demands of an AI-driven economy.

Why it Enhances U.S. Competitiveness: A well-trained AI workforce is essential to maintaining U.S. leadership in AI. By investing in AI education and reskilling, the U.S. can ensure a pipeline of talent capable of developing and deploying AI technologies that will define the future of the global economy.

Priority #4: Strengthening Economic Growth and Job Creation

While AI is often associated with automation and job displacement, a strategic emphasis on AI for societal benefit creates opportunities for new industries and jobs focused on societal improvement. For example, in education, it can lead to the demand for new skill sets and new opportunities such as *curriculum designers* who create AI-augmented learning materials and tools to ensure that education remains relevant in a rapidly evolving technological landscape and *teacher training programs* that prepare educators to incorporate AI-powered tools into classrooms, enhancing their teaching methods and improving student engagement.

Why it Enhances U.S. Competitiveness: A commitment to AI for societal benefit can make the U.S. an attractive hub for global talent and investors. AI technologies that promote effective governance and public services lead to increased overall economic competitiveness. By leveraging AI for social progress, the U.S. not only addresses domestic challenges but also reinforces its global technological and economic dominance.

Conclusion

The University at Buffalo is committed to advancing AI research that has a direct, positive impact on society. As the U.S. continues to shape its AI Action Plan, it is imperative that policy frameworks are crafted with a clear focus on harnessing AI's potential for public benefit while maintaining the nation's leadership in this transformative field.

By investing in AI research, promoting ethical standards, fostering public-private collaborations, and prioritizing workforce development, the U.S. can continue to lead the global AI race. UB is ready to partner with the federal government, industry, and other academic institutions to ensure that AI remains a tool for solving humanity's greatest challenges while reinforcing the United States' position as a global leader in innovation and technology.